MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSEBCA201: INTRODUCTION TO MACHINE LEARNING

MODEL QUESTION PAPER

Duration: 1hr Maximum Marks: 35

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A. Very Short Answer Questions.

Answer Any Five Questions. Each question carries 3 marks.

1. Differentiate between labelled and unlabelled data? [U] [CO1]
2. Explain reinforcement learning with its basic components. [U][CO1]
3. Distinguish between regression and classification. [U][CO1]
4. Define confusion matrix? List its components. [K][CO1]
5. Define logistic regression? State one application. [U[[CO2]
6. State the basic principle of the Naïve Bayes classifier? [U][CO2]
7. What is a perceptron in Artificial Neural Networks? [U][CO2]

(5 X 3= 15 Marks)

Part B. Short Answer Questions.

Answer any 4 Questions. Each question carries 5 marks.

8. Describe the different types of Machine Learning with suitable examples. [U][CO1]
9. Explain the K-Nearest Neighbors (KNN) algorithm. Discuss the impact of the value of k. [A][CO2]
10. Discuss Decision Tree classification. Explain how decision trees are constructed. [An][CO2]
11. Describe Support Vector Machine (SVM) for linearly separable data with a suitable diagram. [U][CO2]
12. Explain the K-Means clustering algorithm. Discuss its advantages and disadvantages. [U][CO2]
13. Discuss the ethical considerations in Machine Learning with suitable real-world examples. [U][CO2]

(4 X 5= 20 Marks)